INTRODUCTION TO HUMAN ANATOMY

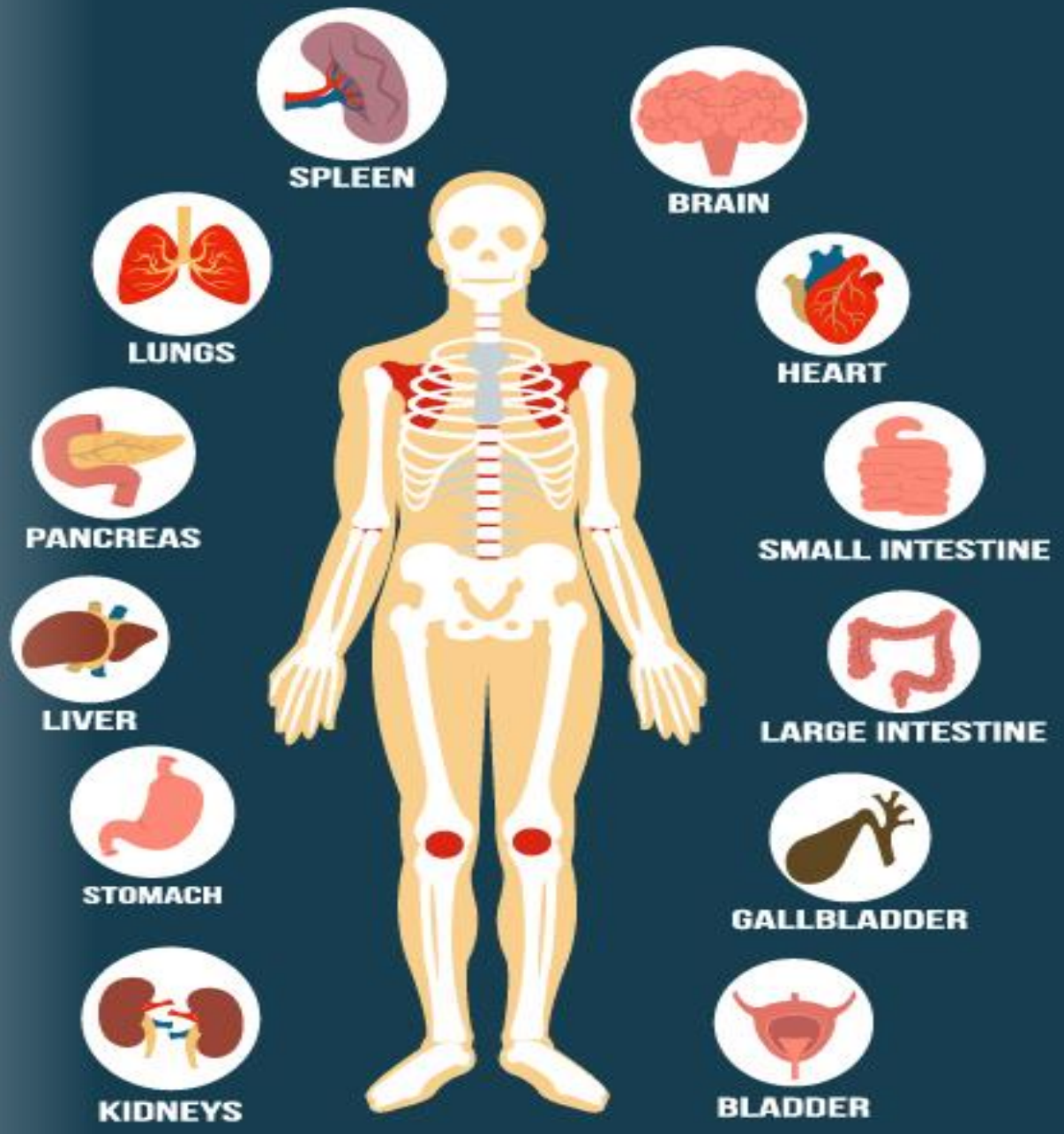
BY

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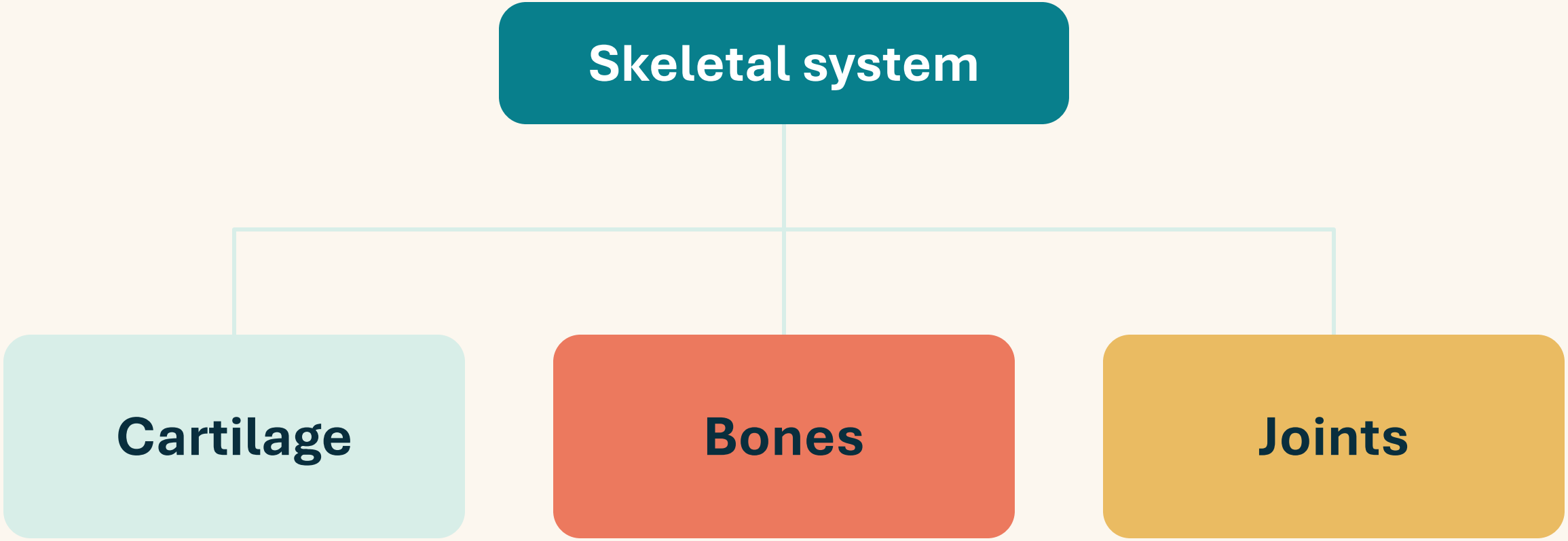
FACULTY OF MEDICINE

MENOUFIA UNIVERSITY



SKELETAL SYSTEM

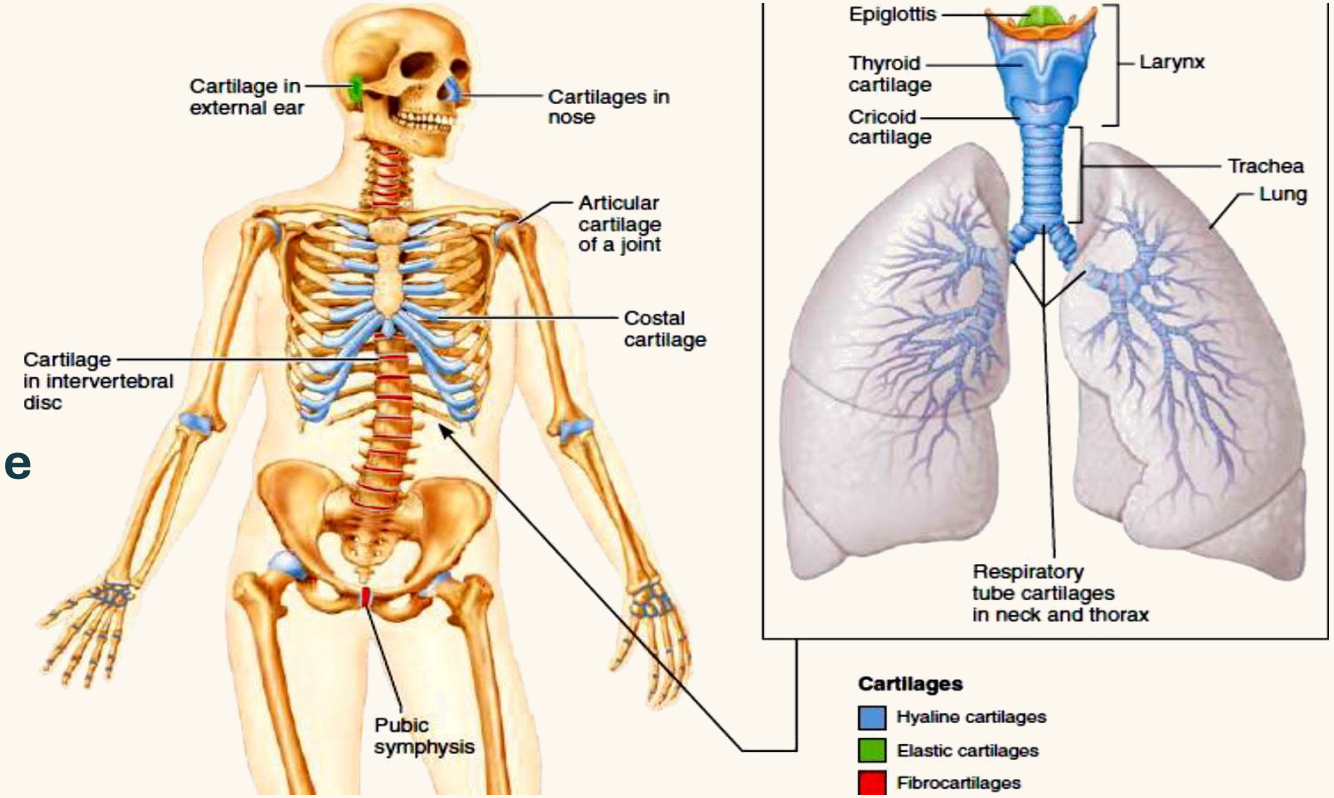



Skeletal system
Cartilage
Bones
Joints

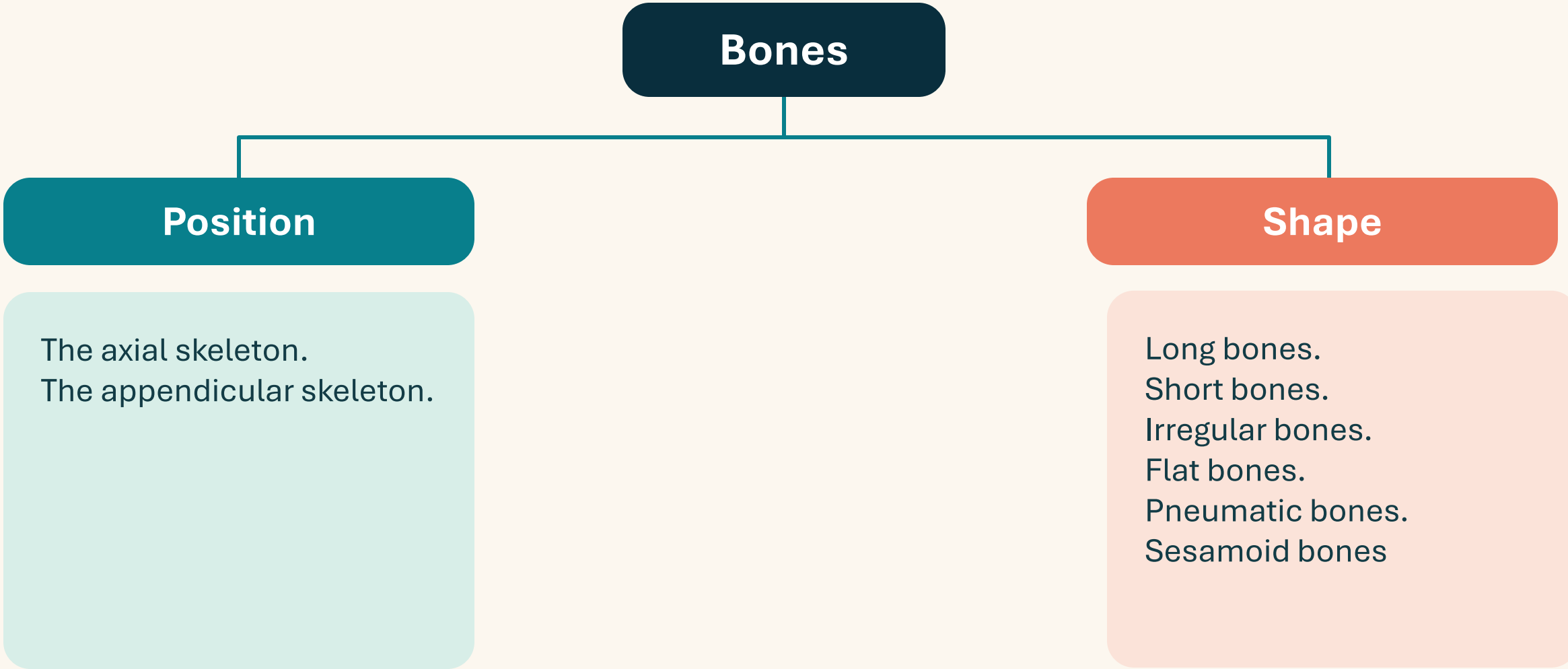
Cartilage

A semi-rigid form of connective tissue that forms parts of the skeleton.

There are three types of cartilage tissue in the body:



Classification of bones according to:



According to position

Axial skeleton

- ✓ **Forms the central axis of the body.**
- ✓ **Includes:**

The skull.
Hyoid: one bone lies in front of the neck
The vertebral column.
Ribs.
Sternum.



Appendicular skeleton

- ✓ **supports the appendages or limbs.**
- ✓ **Includes:**

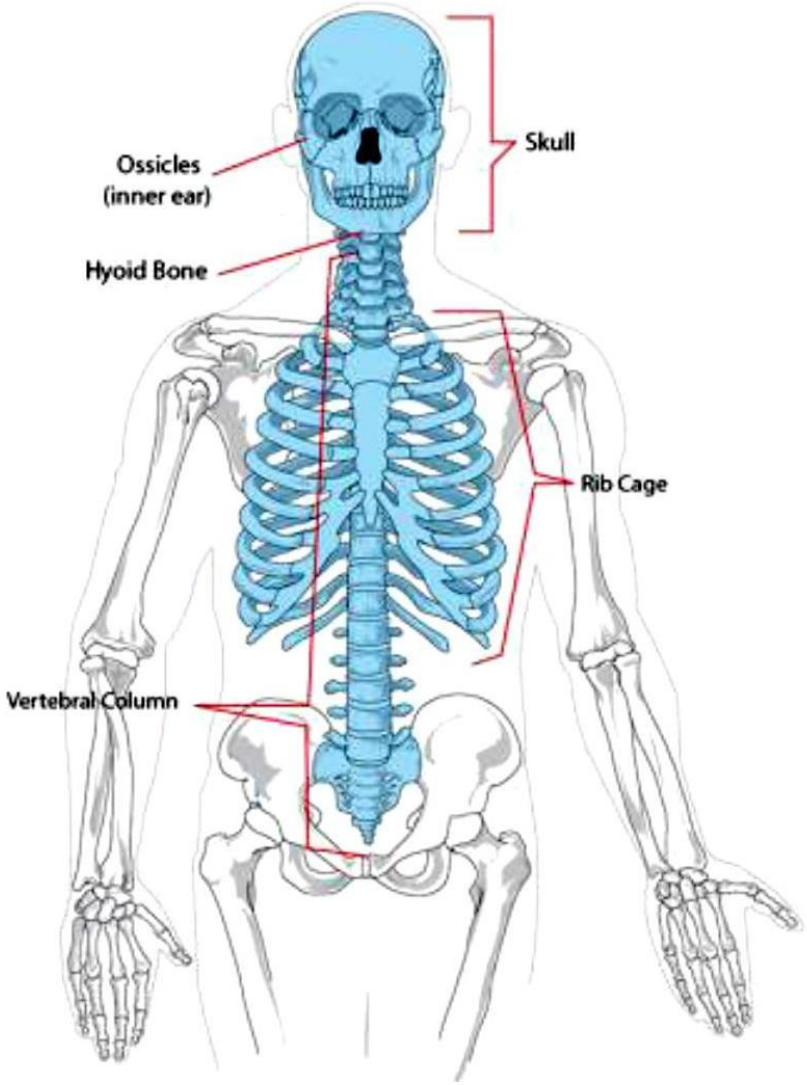
<u>Bones of upper limb:</u>

- The shoulder girdle
- The free upper limb.

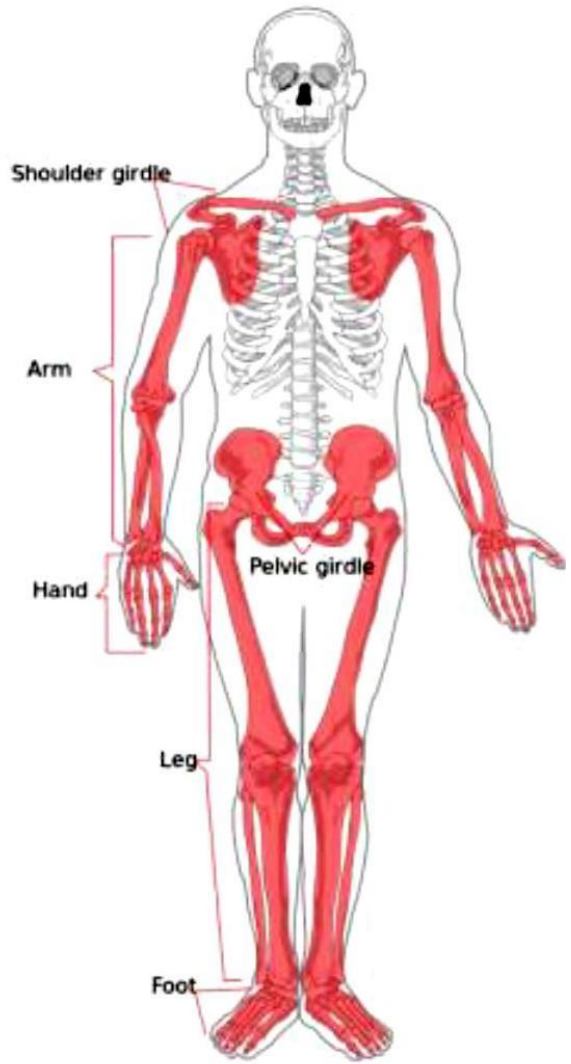
<u>Bones of lower limb:</u>

- The pelvic girdle
- The free lower limb.

AXIAL SKELETON DIAGRAM

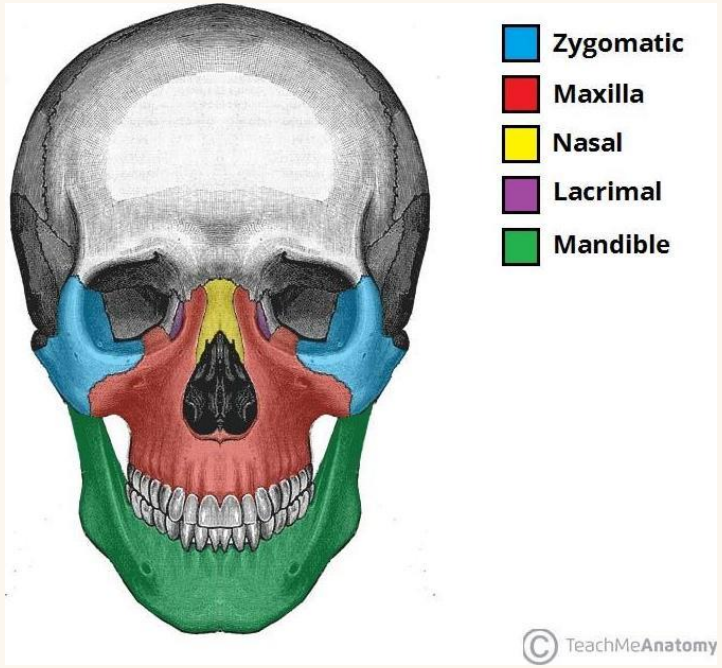


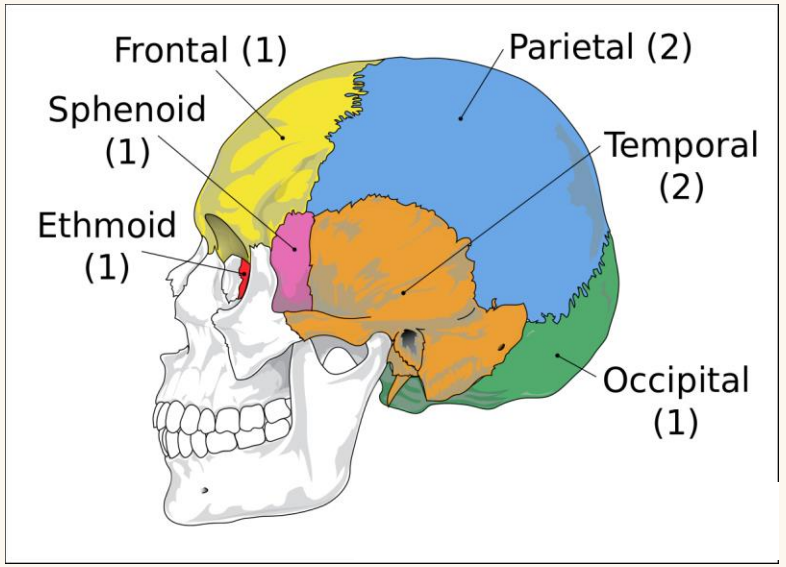
APPENDICULAR SKELETON DIAGRAM



The skull

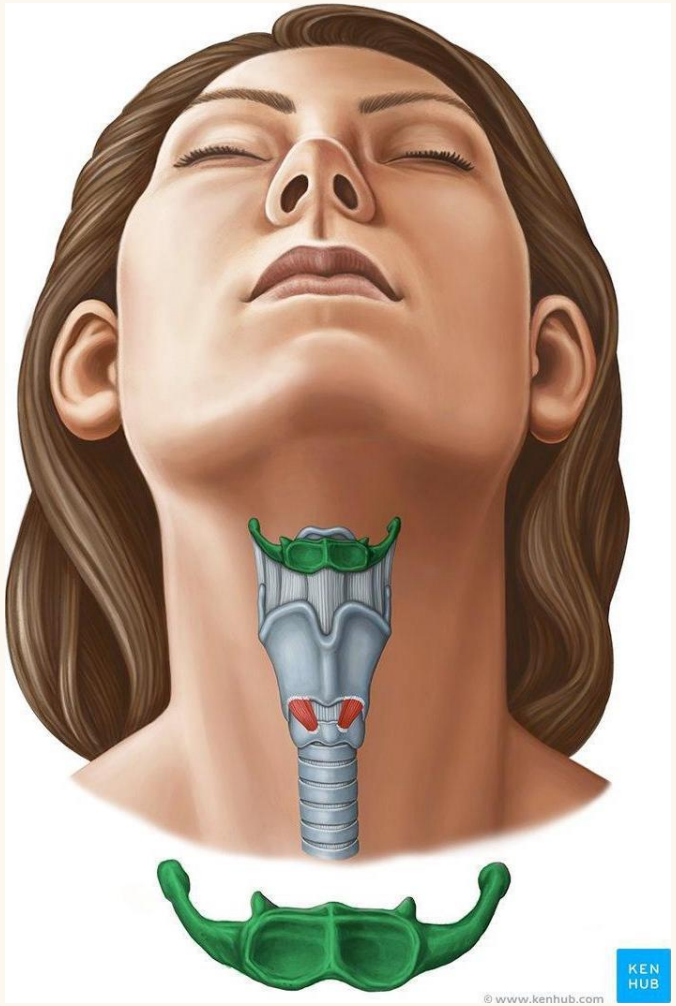
The skull together with the mandible forms the skeleton of the head, but without the mandible, it is called the cranium.





Hyoid bone

Hyoid bone:
one bone lies in front of the neck.

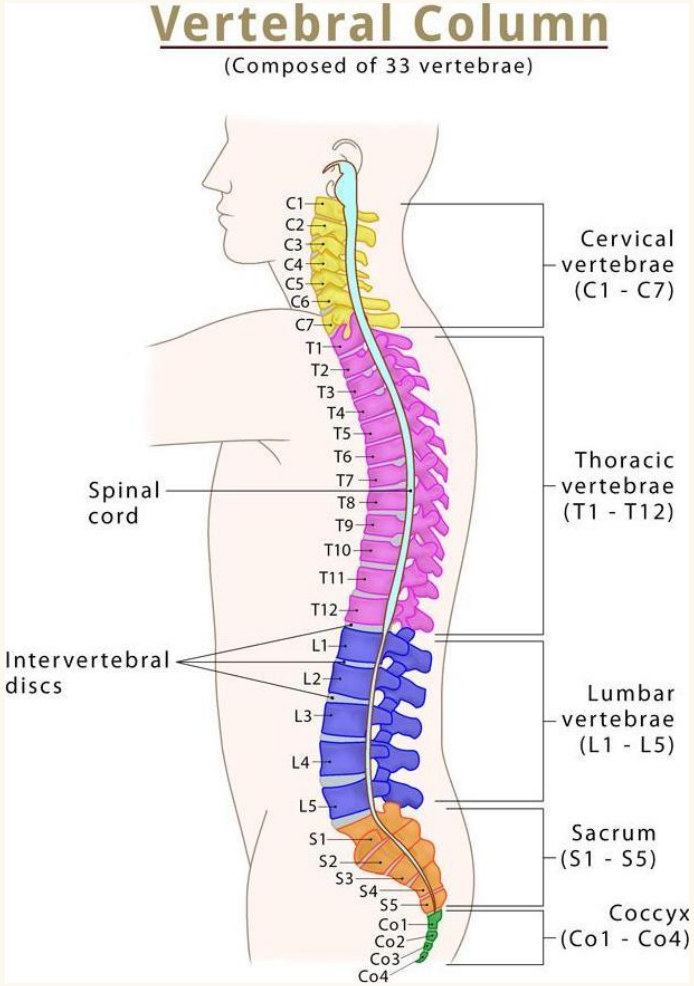


The vertebral column

Formed of a series of bones called vertebrae which articulate with each other by intervertebral discs.

The vertebrae are distributed among five regions as follows:

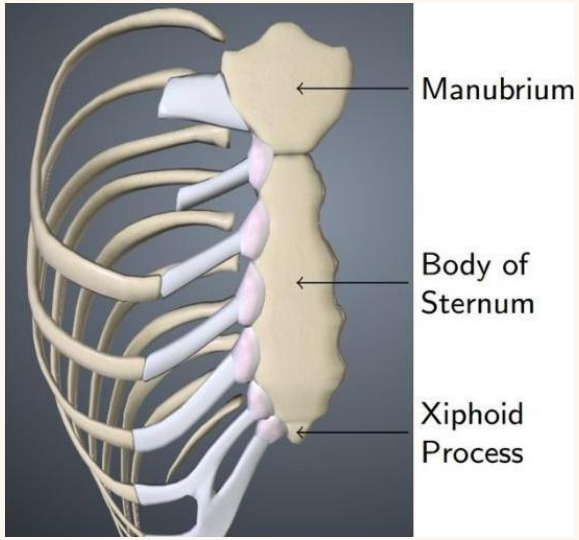
- **7** cervical vertebrae, in the neck region.
- **12** thoracic vertebrae, posterior to the thoracic cavity.
- **5** lumbar vertebrae, supporting the lower back.
- **1** sacrum, consisting of five fused sacral vertebrae.
- **1** coccyx , consisting of four fused coccygeal Vertebrae.



The thoracic cage

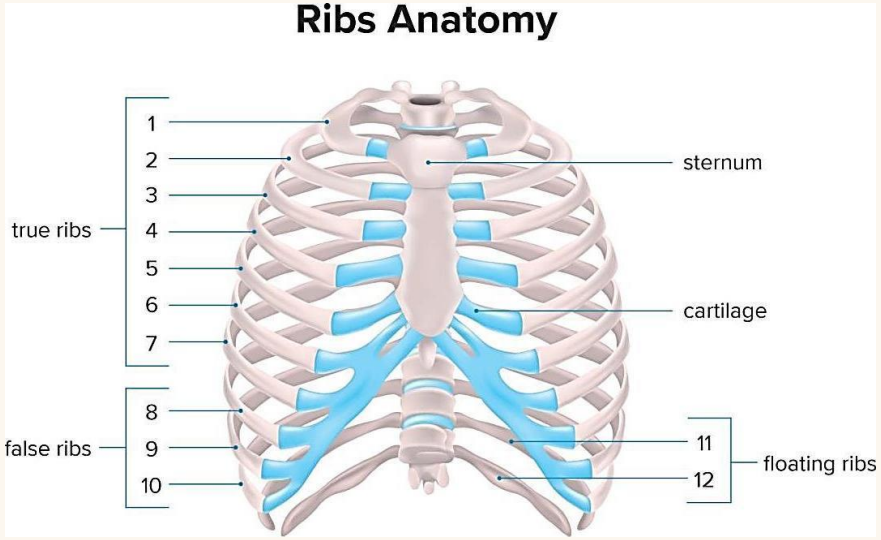
Ribs

12 on each side articulate posteriorly with the thoracic vertebrae and anteriorly with the sternum, forming together the thoracic cage.



Sternum

It is the anterior bone of the thoracic cage which is formed of three parts; manubrium, body and xiphoid process.



The thoracic cage

True ribs: 1-7 ribs

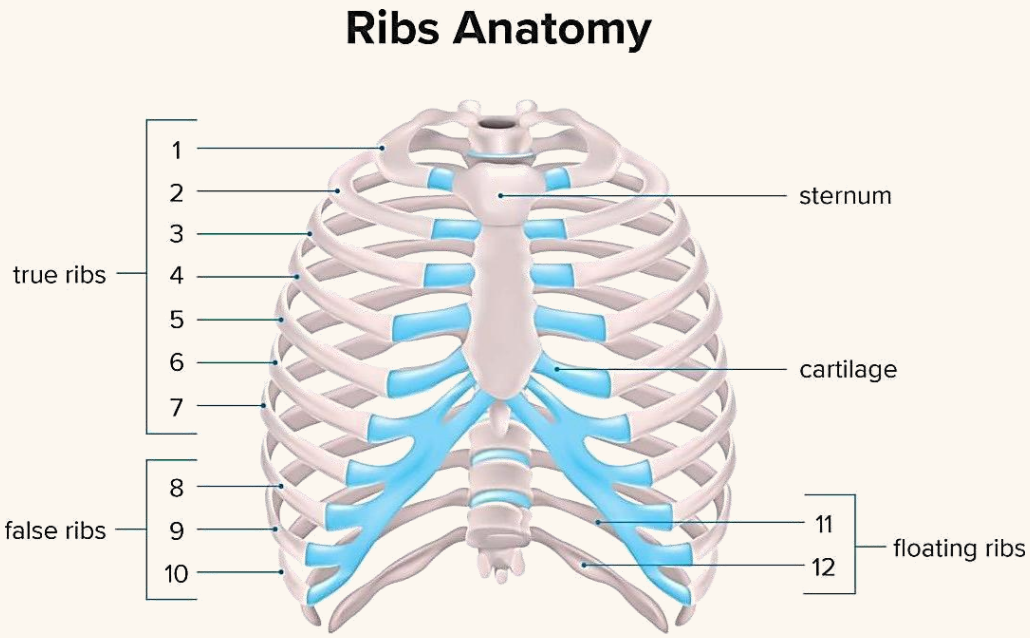
attached to the sternum directly via the costal cartilages

False ribs: 8-10 ribs

are not attached to the sternum directly

Floating ribs: 11 &12 ribs

are not attached to the sternum



Appendicular skeleton

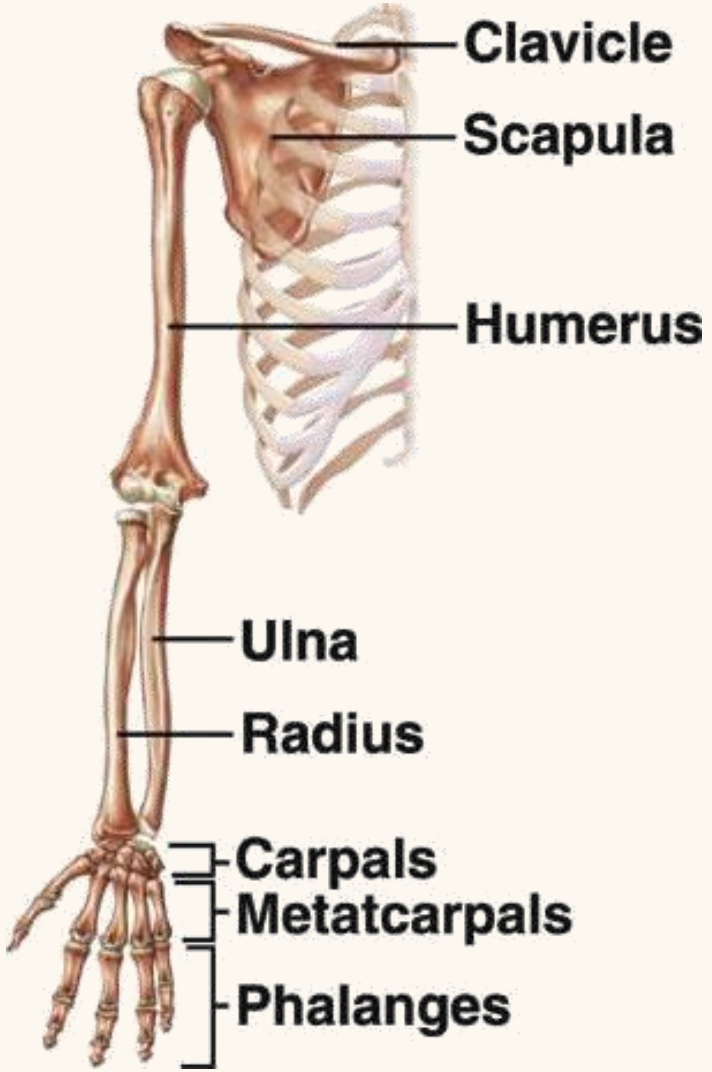
Bones of upper limb

The shoulder girdle: consists of the clavicle anteriorly and the scapula posteriorly

The free upper limb: The three segments of the upper limb are the arm, the forearm, and the hand.

The humerus, is the only bone of the arm

The radius laterally and **the ulna medially** form the skeleton of the forearm.

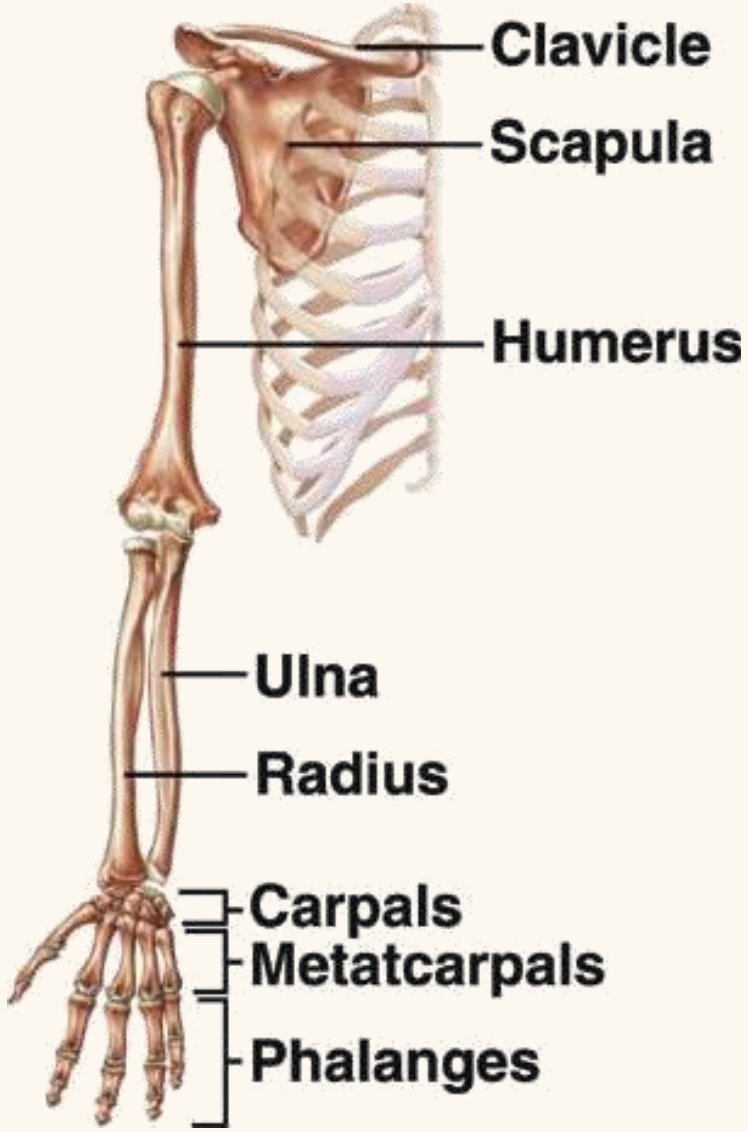


The skeleton of the hand includes:

8 Carpal bones, bones of the wrist (8).

5 Metacarpal bones, the bones of the palm (5).

14 Phalanges, bones of the fingers two for the thumb, and three for each of the fingers. (14).



Appendicular skeleton

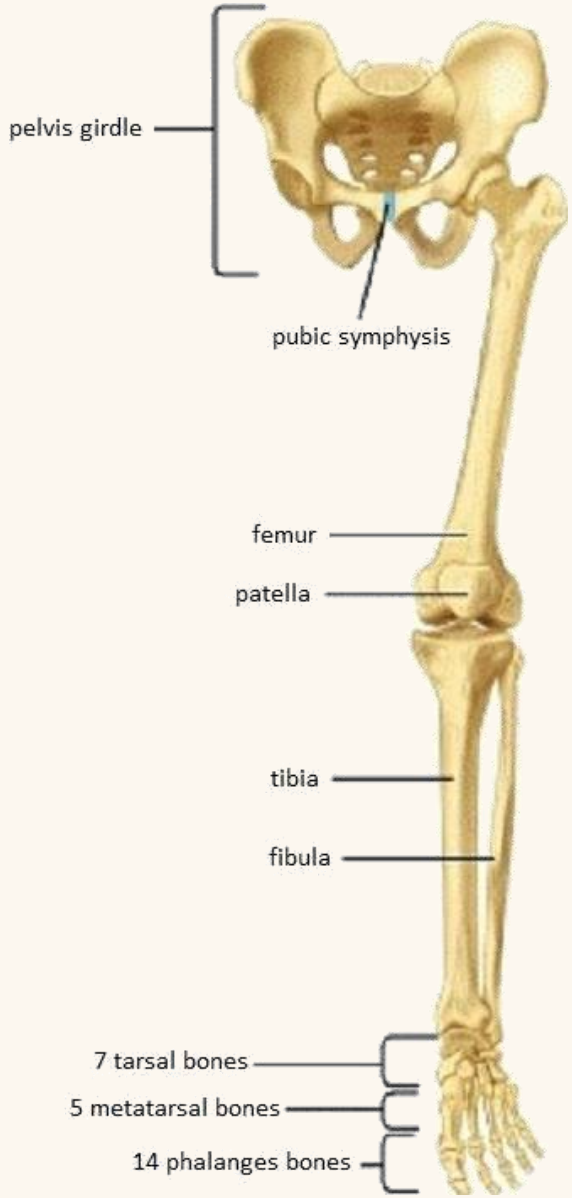
Bones of lower limb

The pelvic girdle: consists of the hip bones.

The free lower limb: The three segments of the upper limb are the thigh, the leg, and the foot.

The femur, is the only bone of the thigh.

The **fibula laterally** and the **tibia medially** form the skeleton of the leg.

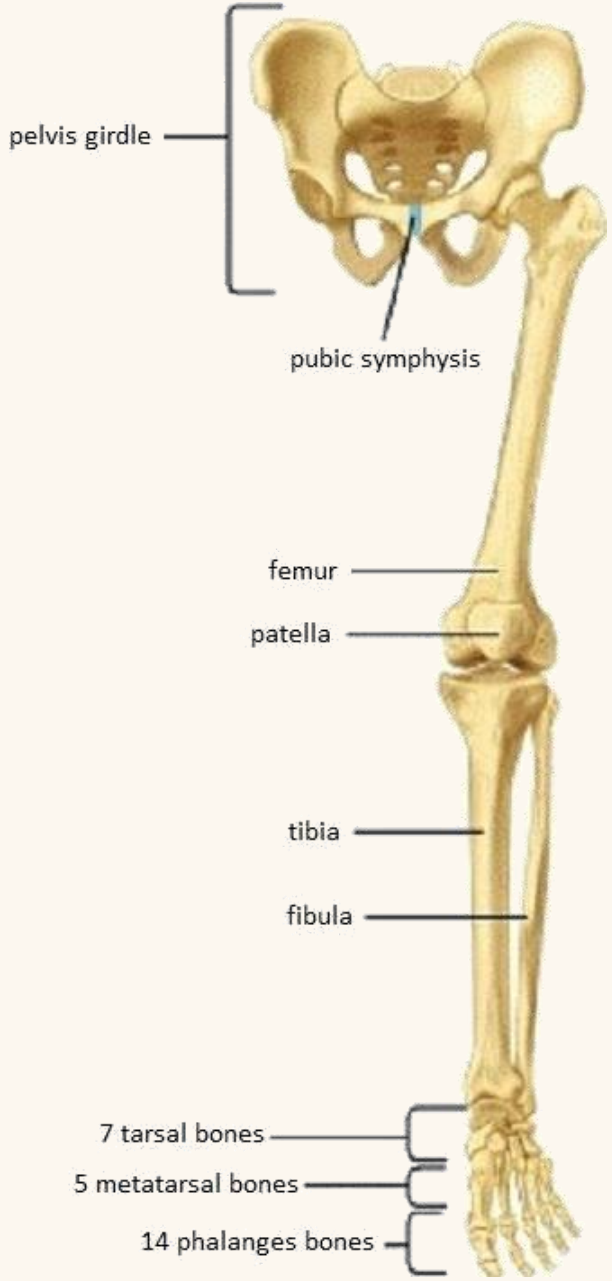


The skeleton of the foot includes:

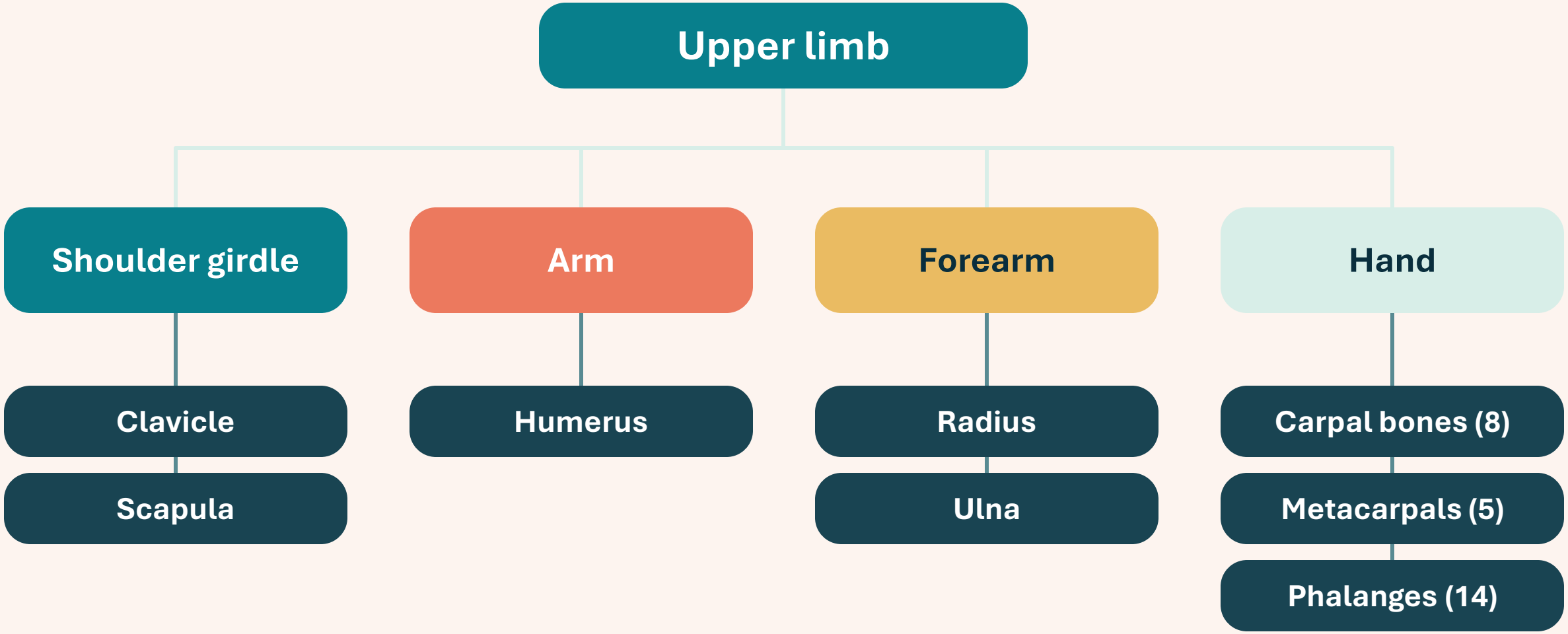
7 Tarsal bones, bones of the ankle (7).

5 Metarsal bones, the bones of the foot (5)

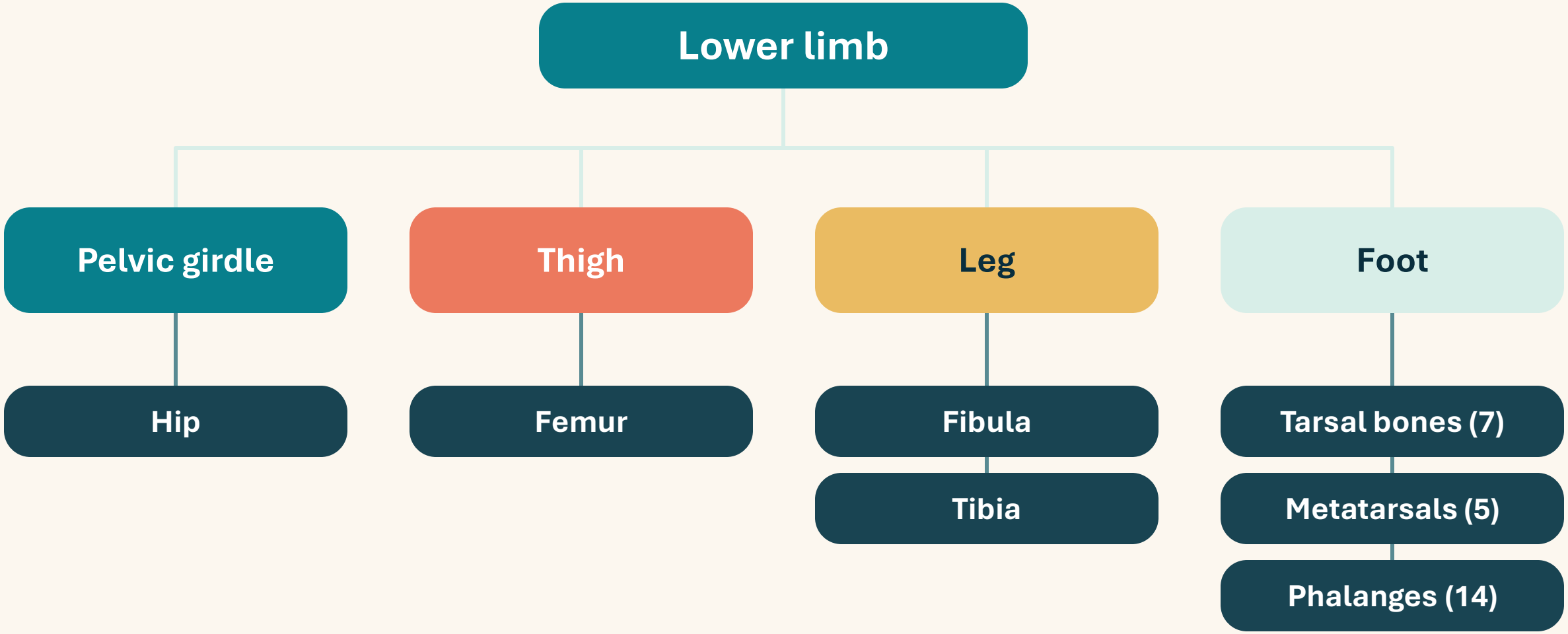
14 Phalanges, bones of the toes, two for the big toe and three for each toe (14).

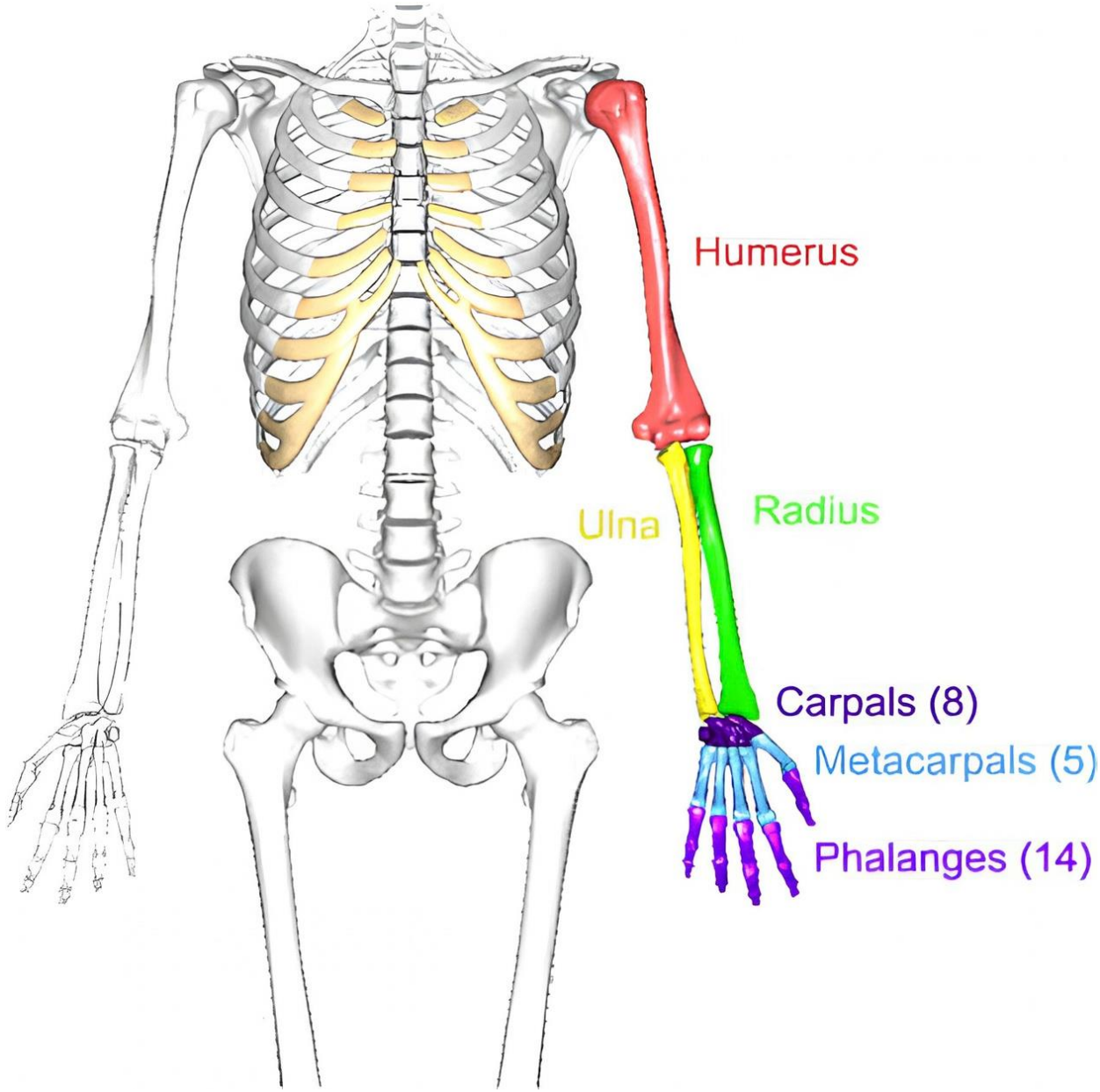


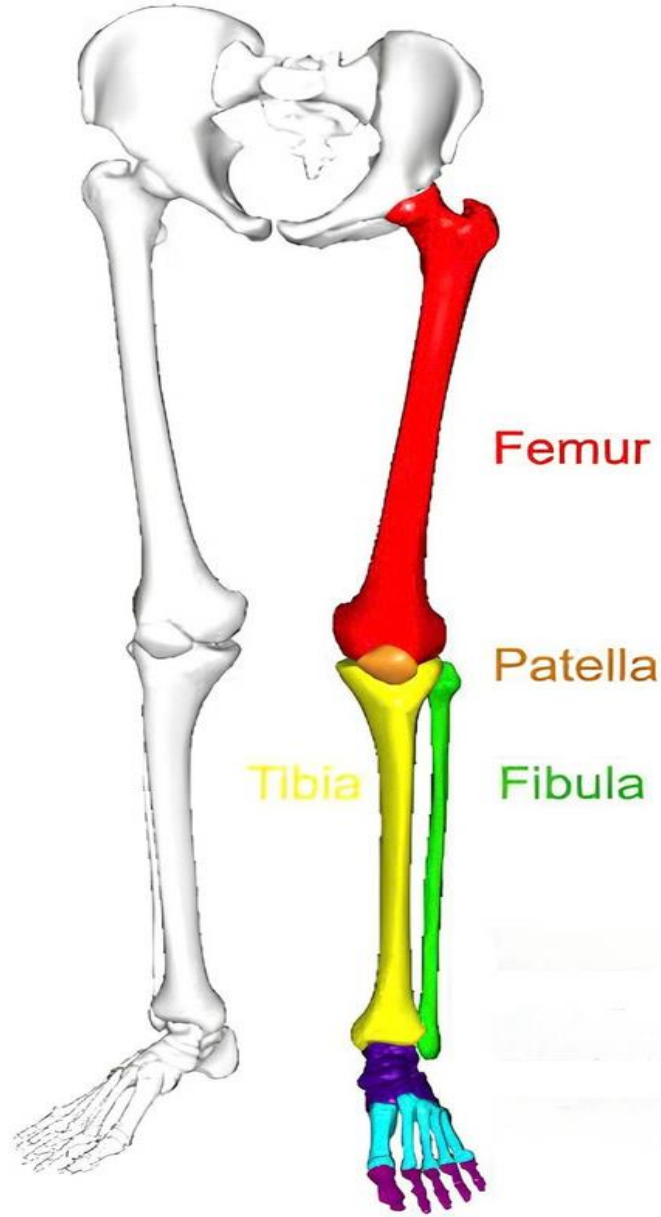
Upper limb


Upper limb
Shoulder girdle
Arm
Forearm
Hand
Clavicle
Scapula
Humerus
Radius
Ulna
Carpal bones (8)
Metacarpals (5)
Phalanges (14)

Lower limb


Lower limb
Pelvic girdle
Thigh
Leg
Foot
Hip
Femur
Fibula
Tibia
Tarsal bones (7)
Metatarsals (5)
Phalanges (14)

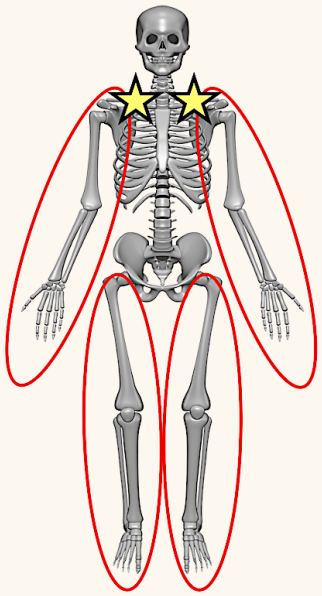

Humerus
Ulna
Radius
Carpals (8)
Metacarpals (5)
Phalanges (14)


Femur
Patella
Tibia
Fibula

Classification of bone according to shape

Long bones:

- ✓ Two ends + shaft.
- ✓ Examples: Bones of the proximal and distal segments of the limbs



Long Bones



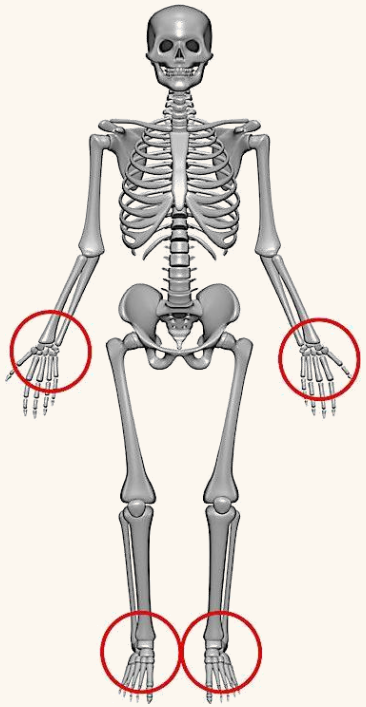
Long* = *Limbs

Long* = *CLavicLe
(CoLLarbone)

Classification of bone according to shape

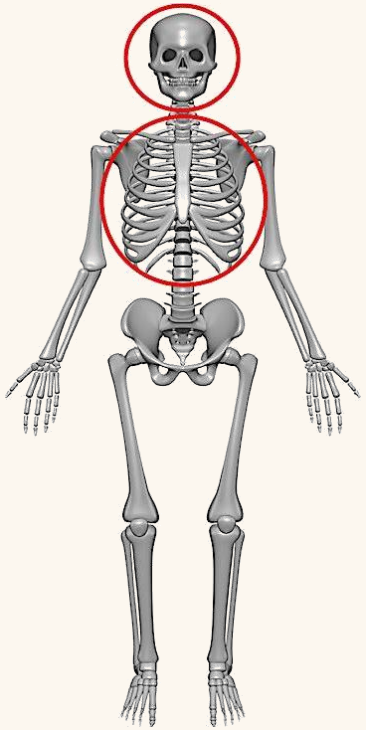
Short bones:

They are generally cubical in shape.
e.g. carpal &tarsal bones.



Flat bones:

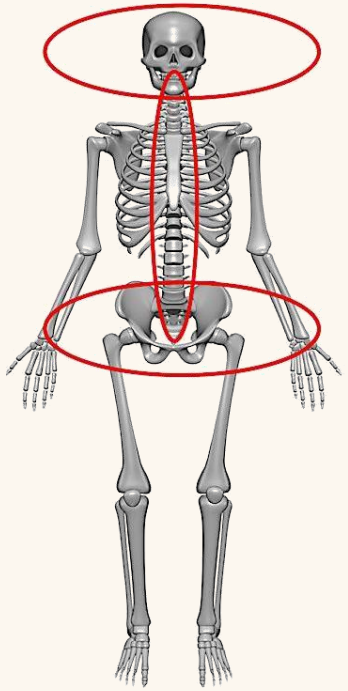
Scapula, ribs, and vault of the skull.



Classification of bone according to shape

Irregular bones:

They are irregular in shape having multiple processes. vertebrae ,hip bone, and bones of the face.





Classification of bone according to shape

Sesamoid bone:

They are small nodules of bone that are found in certain.

The largest sesamoid bone is patella.



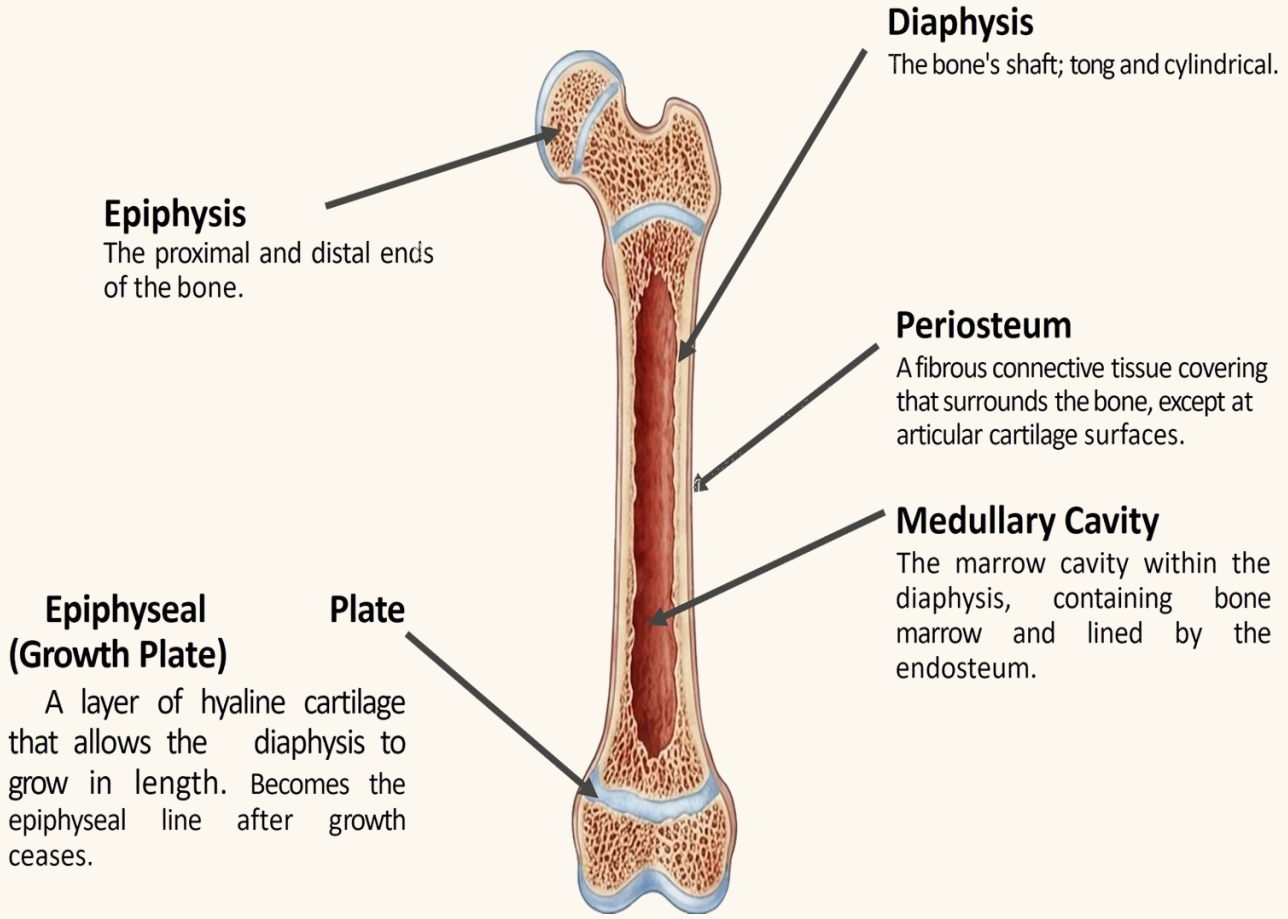
Pneumatic bones:

Skull bones which contain air–filled spaces. e.g. maxilla & frontal bones.



Parts of a long bone

Epiphysis
The proximal and distal ends of the bone.
Diaphysis
The shaft of the bone. Long and cylindrical. Its outer layer is covered by a connective tissue sheath called the **periosteum**.
Medullary Cavity: Also called the marrow cavity.
Lies within the diaphysis.
Contains bone marrow.
Epiphyseal Plate
Also called growth cartilage.
A layer of cartilage.
Allows the diaphysis to grow in length.
When full height is achieved, the cartilaginous epiphyseal plate turns into bone and becomes the epiphyseal line.




Thank
you